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Course Code : CSA0905

1. Design a Class Hierarchy (Pseudocode)

```
START
CLASS Person
    DATA Name, Age
END CLASS

CLASS Student INHERITS Person
    DATA RegisterNo
    DATA Marks[5]
    METHOD CalculateAverage()
        avg <— (Marks[0] + Marks[1] + Marks[2] + Marks[3] +
                                          Marks[4]) /5
        RETURN avg
    END METHOD
END CLASS

CLASS ResearchStudent INHERIT Student
    DATA ResearchArea
    METOD DisplayDetails()
        PRINT Name
        PRINT Age
        PRINT RegisterNo
        PRINT CalculateAverage()
        PRINT Research Area
    END METHOD
END CLASS

BEGIN
Create object RS
Read Name, Age, RegisterNo
```

Read 5 Marks
Read Research Area
RS.DisplayDetails()
END

Sample Input:

Name = Ravi
Age = 20
Register No = 101
Marks = 80 85 90 75 70
Research Area = AI

Output:

Name = Ravi
Age = 20
Register No = 101
Average = 80
Research Area = AI

2. Complete the Missing Pseudocode:

```
BEGIN
READ N
READ ARRAY A

FOR i <— 0 TO N-1
    FOR j <— i+1 TO N-1
        IF A[i] == A[j]
            DELETE A[j]
        ENDIF
    ENDFOR
ENDFOR
```

Sample Input:

N = 6
A = 1 2 3 4 4 5

Sample Output:

Modified Array
1 2 3 4 5

3. Debug the Pseudocode:

```
CLASS Area
METHOD area (length)
RETURN length * length
END METHOD

METHOD are (length, breadth)
RETURN length * breadth
END METHOD

END CLASS
```

Sample Input:

Square side = 5

Rectangle length = 6

Rectangle Breadth = 4

Sample Output:

Square Area = 25

Rectangle Area = 24

4. Trace the Execution

```
sum <— 0
FOR i <— 1 TO 5
    IF i MOD 2 = 0
        sum <— sum + i
    ELSE
        sum <— sum - i
    ENDIF
ENDFOR
PRINT sum
```

Output:

-3

5. Second Largest Element:

```
BEGIN
READ N
READ ARRAY A
largest <— -999
second <— -999
FOR i <— 0 TO N-1
    IF A[i] > largest THEN
        second <— largest
        largest <— A[i]
    ELSE IF A[i] > second AND A[i] ≠ largest THEN
        second <— A[i]
    ENDIF
ENDFOR
PRINT second
END
```

Sample Input:

N = 5

A = 10 20 30 40 50

Output:

Second largest = 40

6. Convert Problem Statement into Pseudocode:

```
INTERFACE Payment
    METHOD pay (amount)
END INTERFACE

CLASS UPI IMPLEMENTS Payment
    METHOD pay (amount)
        PRINT "Payment Successful"
    END METHOD
```

```
END CLASS
BEGIN
    READ amount
    CREATE UPI u
    u.pay (amount)
END
```

Sample Input:

Amount = 1000

Sample Output:

Payment Successful

7. Complete the Pseudocode:

```
CLASS Animal
    METHOD speak( )
        PRINT "Animal"
    END METHOD
END CLASS

CLASS Dog INHERITS Animal
    METHOD speak( )
        PRINT "Barks"
    END METHOD
END CLASS

Animal obj <— NEW Dog( )
obj.speak( )
```

Output:

Barks

8. Constructor:

```
CLASS Employee
DATA id
DATA name

CONSTRUCTOR (id, name)
    this.id <— id
```

```
    this.name ← name
END CONSTRUCTOR
END CLASS
```

Sample Input:

ID = 101
Name = Arun

Output:

Employee Created
ID : 101
Name : Arun

9.

```
CLASS Vehicle
    DATA vehicleNo , ownerName

    METHOD display()
        PRINT vehicleNo
        PRINT ownerName
    END METHOD
END CLASS

CLASS Car INHERITS Vehicle
    DATA fuelType
END CLASS

CLASS Truck INHERITS Vehicle
    DATA Load Capacity
END CLASS

BEGIN
    READ type
    IF type = "Car" THEN
        CREATE Car C
        READ vehicleNo, ownerName, fuelType
        C.display ( )
```

```
    PRINT fuelType
  ELSE
    CREATE Truck T
    READ vehicleNo, ownerName, loadCapacity
    T.display()
    PRINT loadCapacity
  ENDIF
END
```

Sample Input:

Type = Car

Vehicle No = TN01AB1234

Owner Name = Ravi

Fuel Type = Petrol

Sample Output:

Vehicle No : TN01AB1234

Owner Name : Ravi

Fuel Type : Petrol.